

Basic Principles of Medicine 2

Nervous System and Behavioral Sciences

Chemical Senses DLA: Olfaction

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Assigned Reading

- Siegel and Sapru, Essential Neuroscience, 4th edition, © Wolters Kluwer (2019)
- [Chapter 17: Olfaction and Taste](#)



Objectives

A student should be able to:

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0266** Describe different smell qualities and explain the functional anatomy of the olfactory system, focusing on olfactory receptor neurons, the signal transduction process, the olfactory bulb and its circuits
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0267** Describe the olfactory pathways, the location of cortical olfactory areas, and their relation to our emotional system

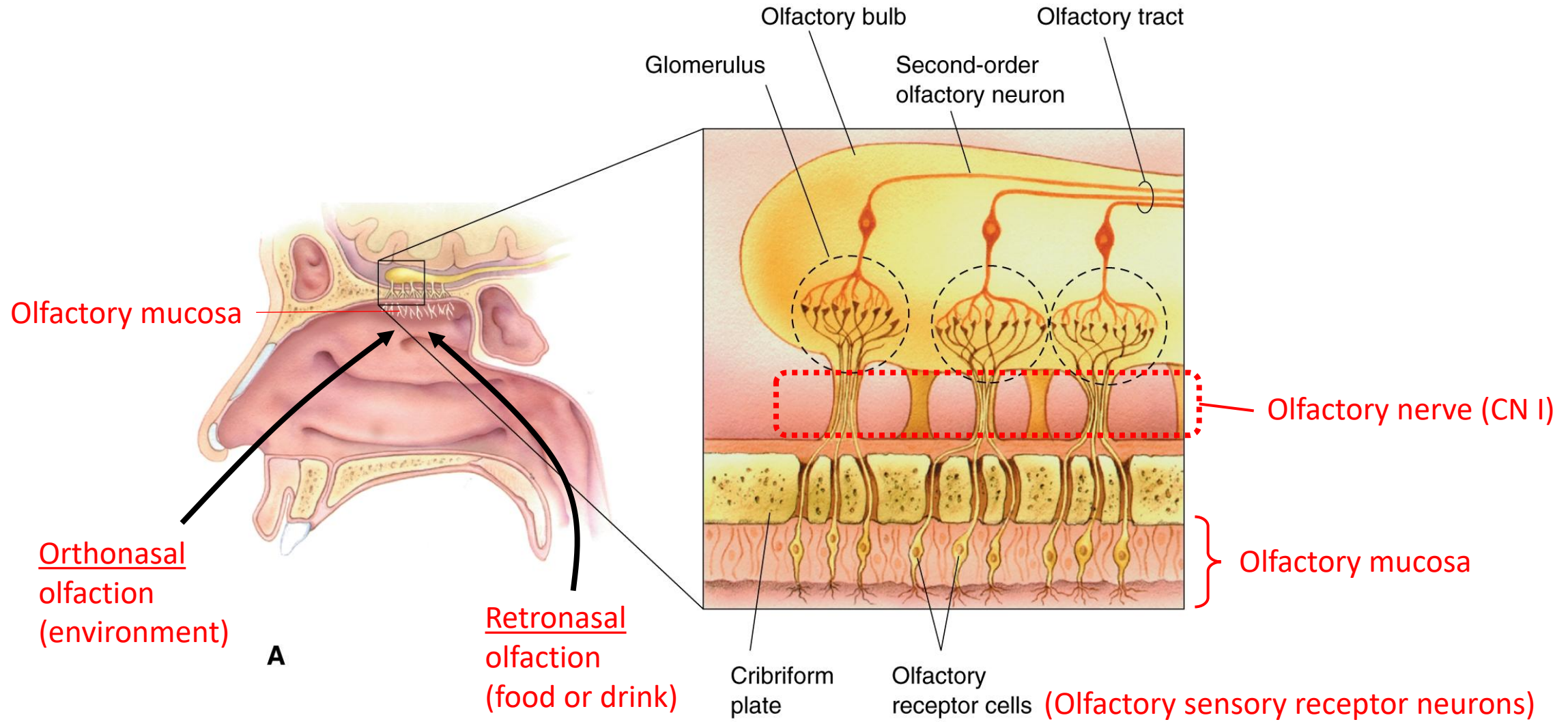


Olfactory Stimuli

- Olfactory stimuli are airborne molecules of our environment, entering the nasal cavity through our nostrils (orthonasal olfaction) or originating from food or drinks in our oral cavity (retronasal olfaction).
- Most odorant molecules stimulate more than just one of the approximately one thousand different types of olfactory receptor neurons. Depending on which sets of receptors are activated, we can differentiate between tens of thousands of different odorants (for example citrus, peppermint, or almond).
- But, in comparison to our sense of smell, we can only differentiate between four or five taste qualities.
- Therefore, our olfactory system contributes very much to the pleasure we experience when eating an elaborate meal.
- In addition, chemoreceptors of the trigeminal somatosensory system contribute to our sensory experience of food, including capsaicin-sensitive heat receptors (red chili pepper), menthol-sensitive cold receptors (mint) and receptors sensitive to mustard oil (wasabi, horseradish) that add pungency.

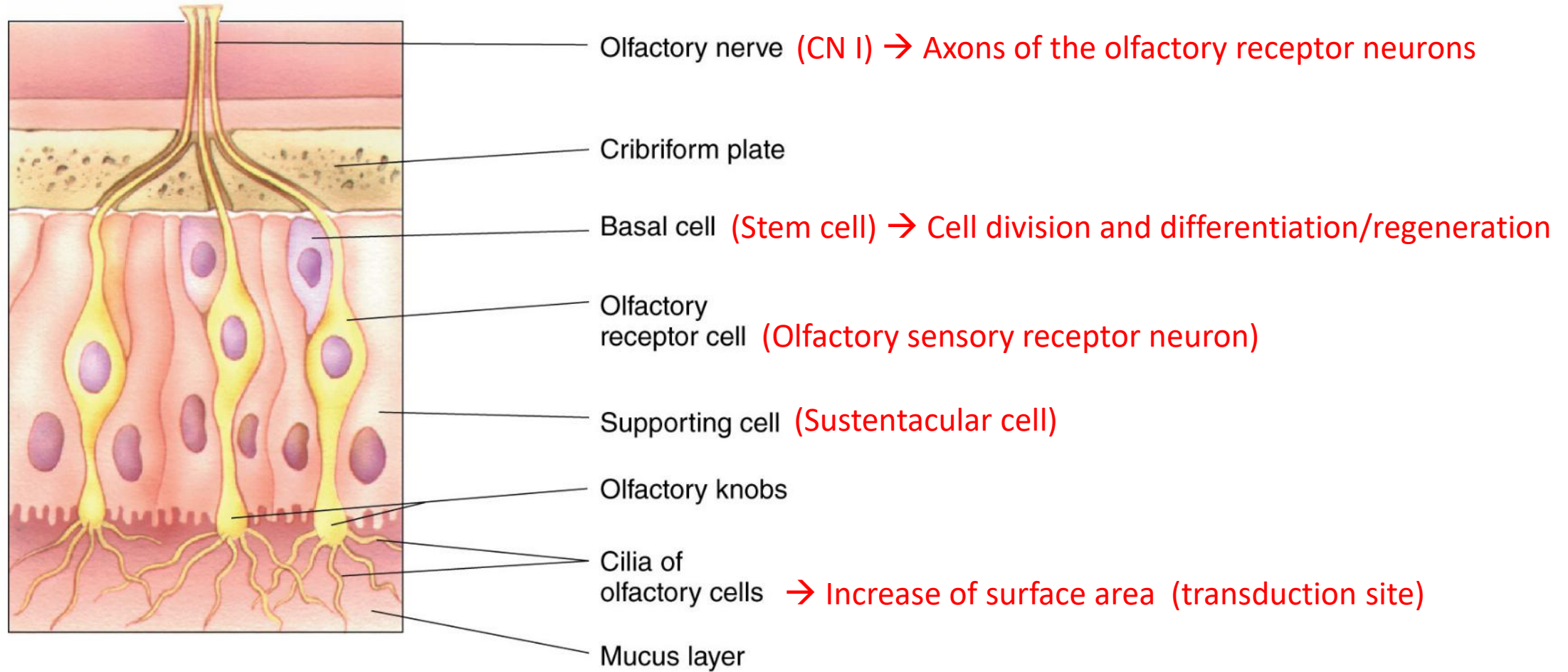


Components of the Olfactory System



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Olfactory Mucosa

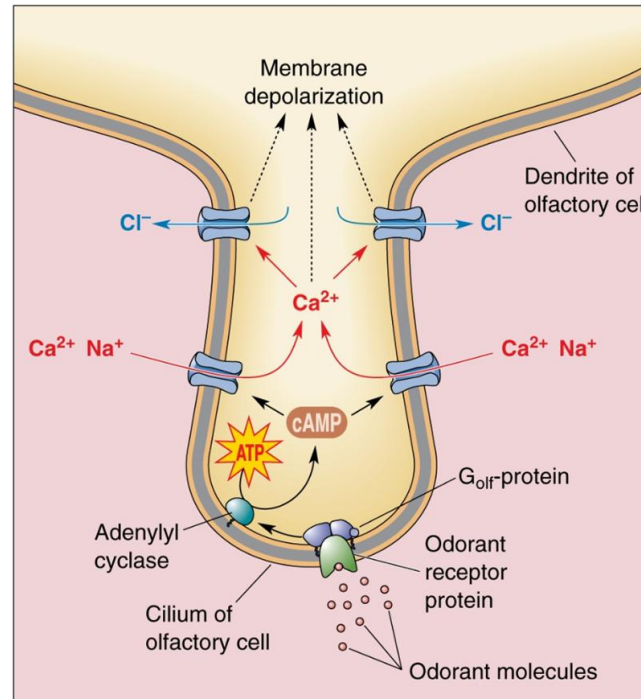


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Olfactory Signal Transduction

- **Metabotropic signal transduction**
 - Second messenger: **cyclic AMP**



Odorant molecule → Receptor binding → G-protein → Enzyme →
Second messenger → Current through ion channel → Depolarization →
Transmitter release

Objectives

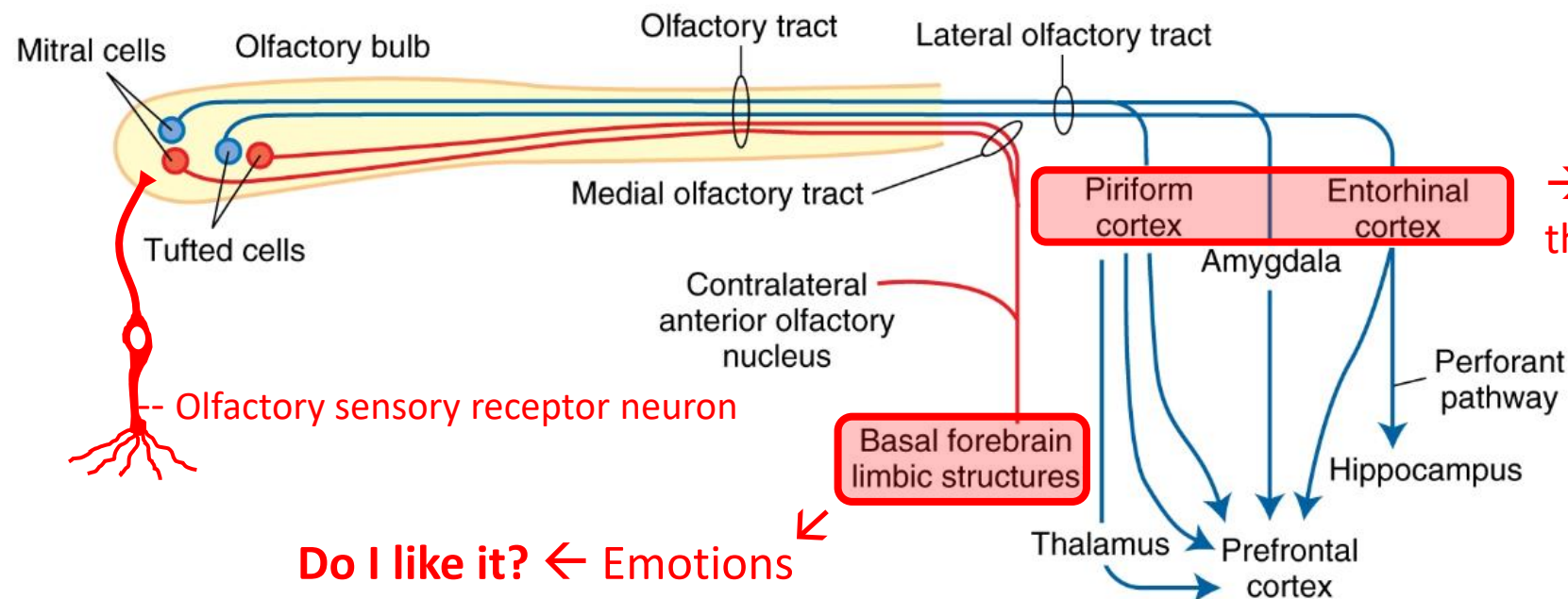
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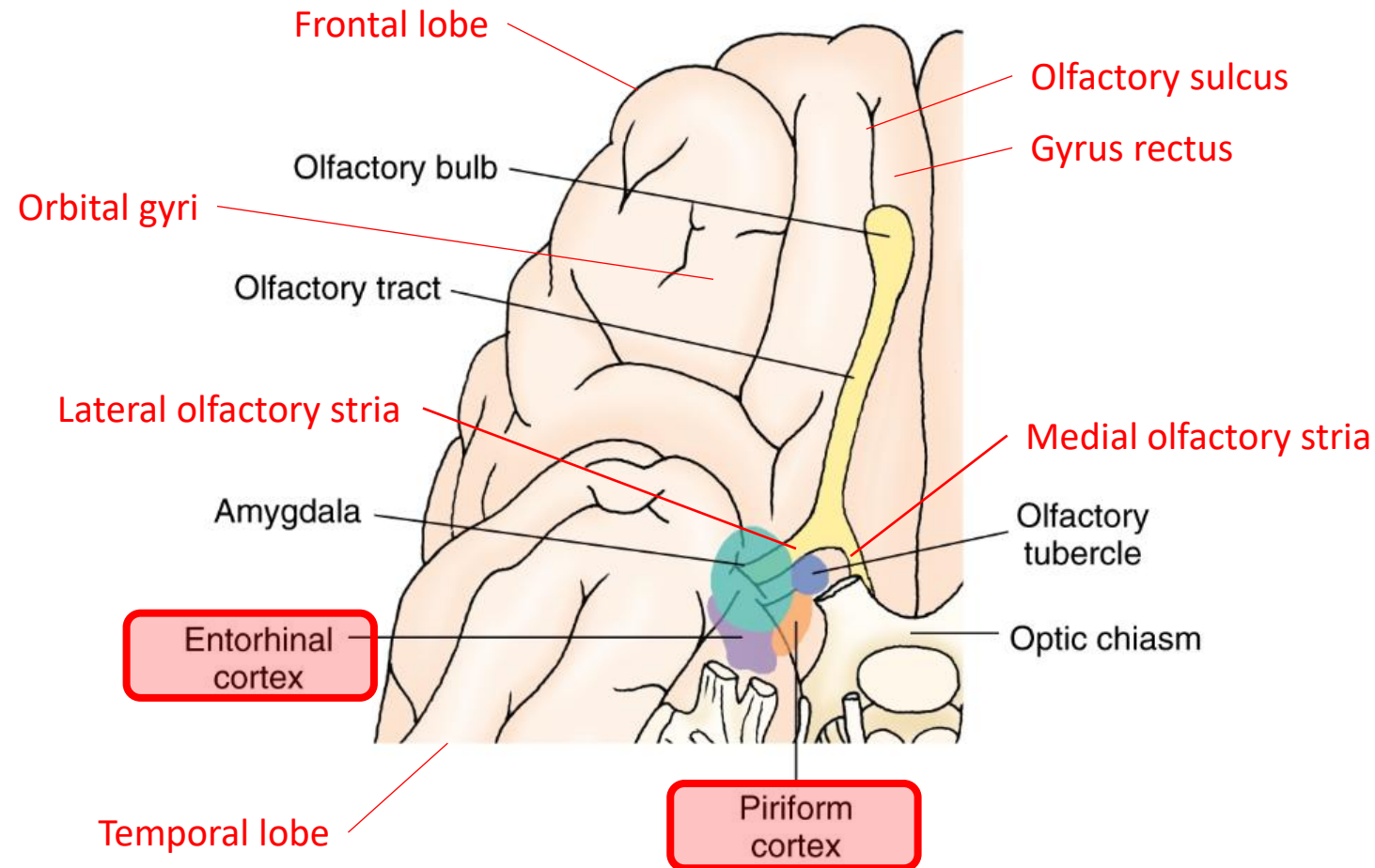
Olfactory Pathways

- The olfactory bulb receives input through the axons of the olfactory receptor neurons forming the olfactory nerve (CN I)
- Output neurons of the olfactory bulb run their axons through the olfactory tract and the medial and lateral olfactory stria. They synapse on structures of the limbic system and the olfactory cortex, without having passed through the thalamus – an exception to the rule for sensory systems, which will become clear as we progress through the module.



Olfactory Cortex

- Olfactory cortex areas in the temporal lobe include the piriform cortex and the entorhinal cortex



The End of Chemical Senses DLA: Olfaction

If you have any questions, please send an email to:

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Dr Noah Leton (GD) <Nleton@sgu.edu>